

STAT GR5291/GU4291 Final Project Report

Dynamic Multi-Asset ETF Allocation with Risk-Sensitive Reinforcement Learning

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This report explores whether a transparent reinforcement-learning allocator can add value in a small multi-asset ETF portfolio. The analysis is based on four liquid ETFs and uses a fully chronological train-validation-test setup. We compare the risk-sensitive RL policy with simple benchmark strategies that are easier to interpret. The results do not point to a clear winner. The RL model ends with the highest terminal wealth in the test period, but inverse-volatility and minimum-variance strategies still provide better Sharpe ratios and stronger drawdown control.

I. Introduction

This project is built around two practical questions. First, can a dynamic reinforcement-learning allocation rule outperform standard portfolio strategies in terms of out-of-sample return? Second, if it does improve returns, is that improvement still convincing after considering risk, drawdown, and trading intensity?

Our earlier work on the project suggested that these questions should not be answered only through in-sample optimization. Instead, the RL strategy needs to be evaluated against transparent and economically interpretable benchmarks. For this reason, we compare the RL allocator not only with a passive equal-weight portfolio, but also with rebalanced risk-based strategies such as inverse-volatility and minimum-variance allocation. We also keep the ETF universe small, which makes the model behavior easier to interpret and discuss.

The motivation for this design also comes from the literature review completed in Homework 2. Kolm and Ritter (2025) argue that RL is especially useful in finance when decisions are sequential and updated through feedback, but they also emphasize that RL should complement, rather than replace, classical portfolio methods. Wang and Liu (2025) show that risk-sensitive reward design can improve RL behavior in volatile market settings, while Tamar, Glassner, and Mannor (2014) provide a tail-risk perspective through CVaR-based optimization. Kim and Lee (2025) highlight the importance of cross-asset adaptation and walk-forward evaluation in non-stationary markets. Hurst, Johnson, and Ooi (2010) further show that simple capital diversification may still leave investors with concentrated risk exposures. Taken together, these papers support our focus on dynamic allocation, explicit risk penalties, and comparison with interpretable benchmark rules.

The four ETFs are selected to represent different roles in the portfolio:

SPY: broad U.S. equity market exposure

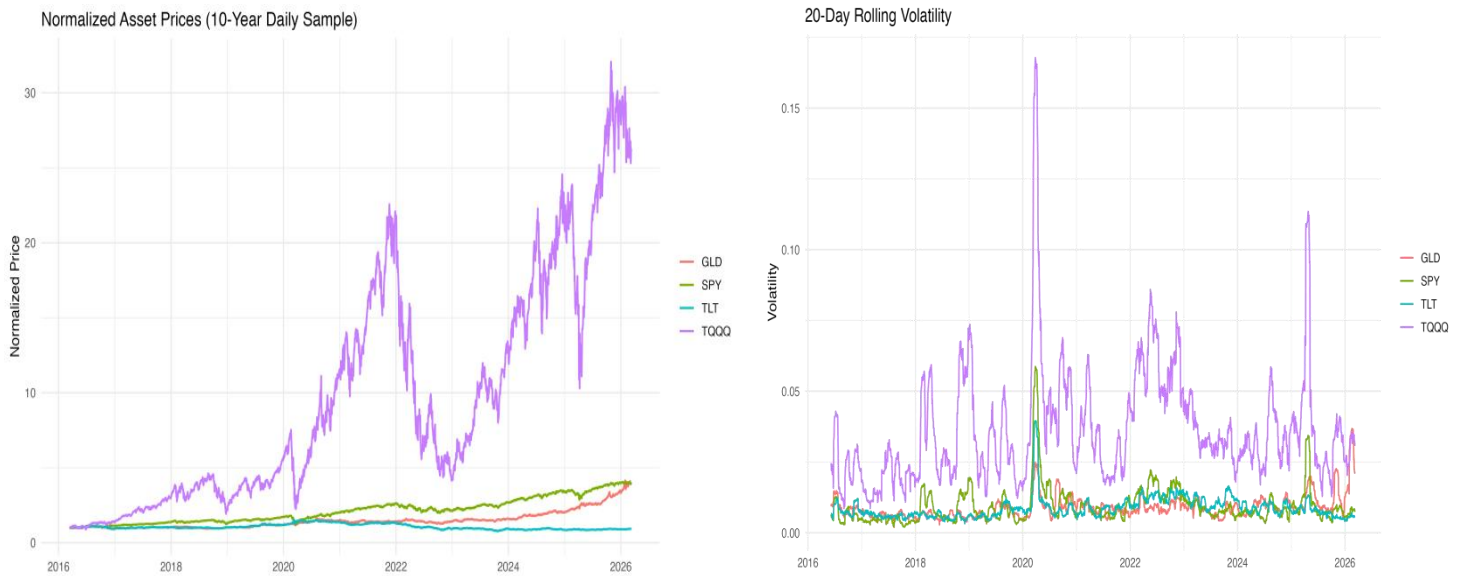
TQQQ: leveraged Nasdaq growth exposure

TLT: long-duration U.S. Treasury exposure

GLD: gold exposure

Figure 1 provides a first look at the behavior of the ETF universe.

Figure 1. Exploratory evidence for the four-ETF universe. Left: normalized ETF price paths over the full sample. Right: 20-day rolling volatility.



The exploratory plots show why this is a meaningful allocation problem. The four assets do not move in the same way over the full sample, and their volatility patterns change noticeably over time. TQQQ has the strongest upside potential, but it also comes with much larger risk. TLT and GLD behave differently across market stress periods, which makes them useful diversifying assets. These patterns suggest that a fixed allocation may be too rigid, and they motivate the use of a dynamic allocation rule.

II. Data Collection and Data Description

The dataset is constructed from four local CSV files of daily adjusted closing prices: *SPY.csv*, *TQQQ.csv*, *TLT.csv*, and *GLD.csv*. After merging the files by trading date, the final synchronized panel contains 2,513 daily observations from March 11, 2016 to March 10, 2026.

We then create rolling return and risk features and match each observation with the next-day return vector. Since the model uses a 60-day rolling window, the first 61 rows are dropped during feature construction. This leaves 2,452 usable observations for the main modeling analysis.

Table 1. Chronological data split used throughout the project.

Split	Start	End	Rows
Train	2016-06-07	2022-04-07	1,471
Validation	2022-04-08	2024-03-21	490
Test	2024-03-22	2026-03-09	491

The chronological split is important because the project is designed as a prediction and allocation exercise, not an in-sample fitting exercise. The training period uses the earliest 60% of the data. The validation period uses the next 20% to choose reward coefficients and tune model settings. The final 20% is kept as a one-time out-of-sample test set. This setup helps avoid using information from the test period during model selection.

III. Statistical Models

We compare the proposed RL allocation strategy with four simple benchmark rules:

1. Buy-and-hold equal-weight allocation
2. Monthly rebalanced equal-weight allocation
3. Monthly inverse-volatility allocation
4. Monthly minimum-variance allocation with long-only constraints

The benchmark portfolios are intentionally kept simple. Each monthly rebalanced benchmark uses a 60-day lookback window and the same proportional transaction-cost assumption as the RL strategy. This allows the comparison to focus on the allocation rule itself rather than on inconsistent trading assumptions.

The RL model uses a linear stochastic policy. On day t , the state vector combines market information with current portfolio information:

$$\mathbf{s}_t = [\mathbf{x}_t, \mathbf{w}_{t-1}, V_t, \mathbf{DD}_t]$$

Here, $\mathbf{x}_t$ contains 1-day, 5-day, 20-day, and 60-day returns for each asset, together with 20-day and 60-day rolling volatility, 20-day downside deviation, 60-day rolling drawdown, and 20-day pairwise correlations. The state also includes the previous portfolio weights $\mathbf{w}_{t-1}$, current portfolio value V_t , and current drawdown $\mathbf{DD}_t$. All predictors are standardized using only the training-sample means and standard deviations.

The policy first maps the state vector into action logits:

$$\boldsymbol{\mu}_t = \mathbf{W}^\top \mathbf{s}_t + \mathbf{b}, \quad \mathbf{a}_t = \boldsymbol{\mu}_t + \sigma \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

The action logits are then converted into long-only portfolio weights using a softmax transformation, followed by clipping and renormalization:

$$\mathbf{w}_t = \Pi(\text{softmax}(\mathbf{a}_t))$$

Here, the projection operator imposes a 70% single-asset cap. This prevents the strategy from concentrating the entire portfolio in one ETF.

After rebalancing, the portfolio return is computed as:

$$\mathbf{r}_{t+1}^p = \mathbf{w}_t^\top \mathbf{r}_{t+1} - c \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1$$

The training reward is defined as:

$$R_t = \mathbf{r}_{t+1}^p - \lambda (\mathbf{r}_{t+1}^p)^2 + \phi DD_{t+1} - \eta TO_t$$

Where

$$TO_t = \|\mathbf{w}_t - \mathbf{w}_{t-1}\|_1$$

This reward structure encourages higher portfolio returns while penalizing unstable returns, drawdowns, and excessive trading. The policy parameters are updated using a REINFORCE-style policy-gradient procedure, which keeps the learning rule relatively explicit and interpretable.

IV. Results and Research Question

The main question is whether the RL strategy justifies its additional complexity during the held-out test period from March 22, 2024 to March 9, 2026. Table 2 compares the RL strategy with the four benchmark allocation rules using return, risk, drawdown, and turnover measures.

Table 2. Held-out test-period performance comparison.

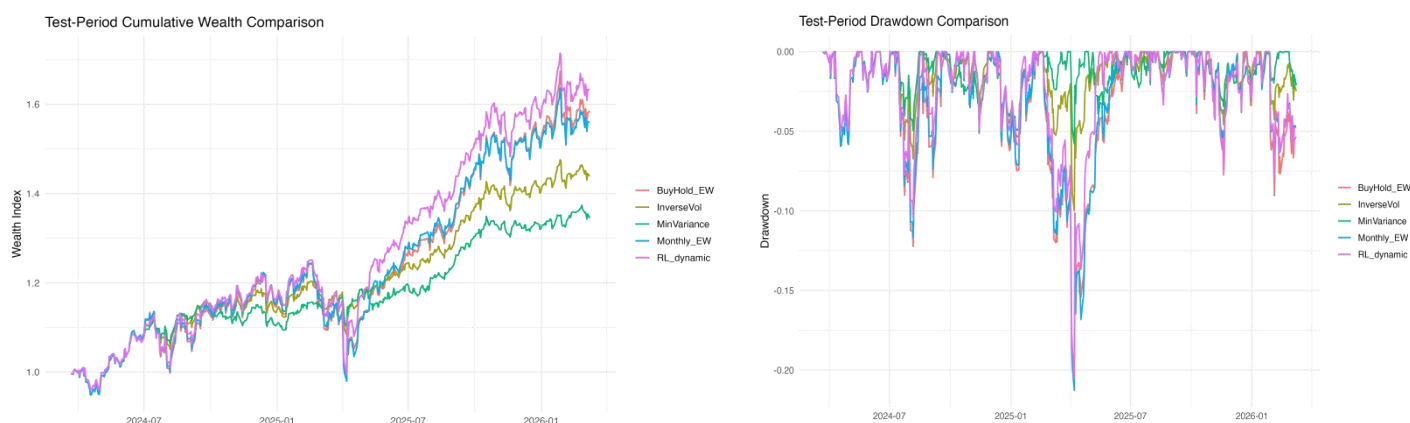
Strategy	Total Return	Ann. Return	Ann. Vol	Sharpe	Max DD	Avg. Turnover
RL	63.3%	28.6%	20.6%	1.32	-20.5%	5.03%
Buy-Hold EW	58.4%	26.6%	20.8%	1.24	-20.6%	1.10%
Monthly EW	56.0%	25.6%	20.5%	1.21	-21.3%	1.19%
Inverse Vol	43.9%	20.5%	12.9%	1.51	-10.0%	1.18%
Min Variance	34.5%	16.4%	10.8%	1.46	-5.9%	1.59%

Table 2 shows that the RL strategy achieves the highest total return and annualized return over the test period. Its total return is 63.3%, compared with 58.4% for the buy-and-hold equal-weight portfolio and 56.0% for the monthly rebalanced equal-weight portfolio. This suggests that the RL policy is able to capture some additional upside during the out-of-sample period.

However, the risk-adjusted results are more mixed. The inverse-volatility portfolio has the highest Sharpe ratio, at 1.51, while the minimum-variance portfolio has the smallest maximum drawdown, at -5.9%. By comparison, the RL strategy has a Sharpe ratio of 1.32 and a maximum drawdown of -20.5%, which are close to the equal-weight benchmarks. The RL strategy also has the highest average turnover, at 5.03%, reflecting the cost of a more active allocation rule.

Overall, the results support a balanced conclusion. The RL strategy performs well in terms of terminal wealth, but it does not clearly dominate the simpler benchmark portfolios once risk, drawdown, and trading intensity are considered. This means that the benefit of RL is mainly visible in return performance, while the risk-based benchmark rules remain more attractive from a risk-control perspective.

Figure 2. Held-out test-period performance. Left: cumulative wealth. Right: drawdown.



The results are favorable to RL in terms of return, but not across all performance dimensions. The RL strategy produces the highest total return, 63.3%, and the highest annualized return, 28.6%. However, it does not deliver the best risk-adjusted performance. The inverse-volatility strategy has the highest Sharpe ratio, at 1.51, while the minimum-variance strategy has the smallest maximum drawdown, at about -5.9%. RL also trades more actively, with average turnover slightly above 5.0%, compared with roughly 1.1% to 1.6% for the benchmark rules.

Overall, the test results are mixed: RL improves raw return and terminal wealth, but its advantage weakens once Sharpe ratio, drawdown control, and trading intensity are considered.

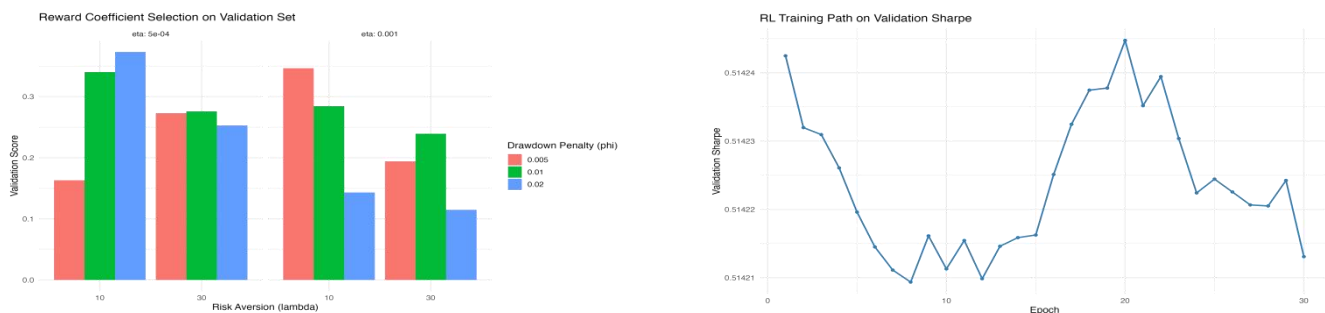
V. Model Validation and Generalization

Model selection is based only on the validation window, not on the test period. We search over $\lambda \in \{10, 30\}$, $\phi \in \{0.005, 0.01, 0.02\}$, and $\eta \in \{0.0005, 0.001\}$, while keeping the transaction cost, exploration noise, and maximum-weight cap fixed.

The validation criterion is designed to balance risk-adjusted return and drawdown control:

$$\text{ValidationScore} = \text{Sharpe} - 0.5 \times |\text{MaxDrawdown}|$$

Figure 3. Validation diagnostics for the RL model. Left: reward-grid search by validation score. Right: validation Sharpe across epochs for the selected specification.



The validation results show that the selected specification is not simply the one with the highest raw validation return. It is chosen using a return-drawdown tradeoff and is then carried forward unchanged to the test period, which helps keep the final comparison genuinely out of sample.

The validation exercise is intentionally modest. The grid search covers only 12 parameter combinations, so it should be interpreted as a structured sensitivity check rather than proof of global optimality. The top-ranked settings cluster around $\lambda=10$, which suggests that the result is fairly stable within the tested grid. Still, because the model is evaluated on one chronological split and one training run, some validation-set overfitting cannot be ruled out.

Generalization is encouraging but not perfect. On the held-out test window, the RL strategy keeps its return advantage, suggesting that the policy is not purely fitted to the training period. However, the COVID stress test shows a limitation: RL slightly outperforms buy-and-hold in total return, but experiences a deeper drawdown. This suggests that the model can adapt, but not always quickly enough during abrupt regime changes.

As an additional robustness check, we use a moving-block bootstrap on the daily test-period returns, with 20-trading-day blocks and 2,000 replications. The bootstrap focuses on the two main comparisons: RL versus buy-and-hold equal weight in total return, and RL versus inverse-volatility in Sharpe ratio.

Table 3. Block-bootstrap significance checks on the held-out test window.

Comparison	Metric	Observed	95% CI	Tail Prob	95% CI Excludes 0
RL vs Buy-Hold EW	Total return difference	4.9%	[-9.8%, 19.9%]	0.489	No
RL vs Inverse Vol	Sharpe difference	-0.18	[-0.71, 0.38]	0.408	No

The bootstrap results make the headline ranking less decisive. Although RL has a 4.9% observed total-return advantage over buy-and-hold equal weight, the 95% confidence interval ranges from -9.8% to 19.9%, so zero remains plausible. Similarly, the observed Sharpe difference between RL and inverse-volatility is -0.18, and its 95% interval also includes zero.

Overall, RL reaches the highest realized wealth, but its advantage is not statistically reliable under the bootstrap check. The results are therefore suggestive rather than conclusive, and future work should focus on walk-forward testing, repeated seeds, stress-period analysis, and turnover control.

VI. Conclusion

This project finds that a transparent RL portfolio allocator can improve return performance in a simple four-ETF setting. On the held-out test period, the RL strategy produces the highest terminal wealth and annualized return.

However, RL does not clearly dominate the traditional benchmarks. Inverse-volatility and minimum-variance portfolios remain stronger in Sharpe ratio, drawdown control, and turnover. Overall, RL appears useful as a return-seeking extension to classical allocation methods, but its value should be judged against simple and robust benchmark strategies.

VII. Appendix

Literature Review and Connection to the Project

The Homework 2 literature review directly informed the design of this report and the modeling choices in the project. Table 4 summarizes the five papers selected by the group and the way each one connects to the final project.

Table 4: Homework 2 literature review distilled into the themes used in the final project.

Paper	Contribution	Connection
Wang & Liu (2025)	Shows that drawdown- and volatility-aware reward design can improve risk-adjusted RL behavior.	Motivates our risk-sensitive reward rather than pure return maximization.
Kim & Lee (2025)	Demonstrates that multi-asset RL can adapt to non-stationary market structure and benefits from walk-forward testing.	Supports our multi-asset setting and points to stronger future validation design.
Kolm & Ritter (2025)	Reviews RL applications in trading, hedging, and portfolio choice and argues RL should complement classical models.	Matches our conclusion that RL is useful, but not a full replacement for traditional allocation.
Tamar, Glassner, & Mannor (2014)	Provides a direct way to optimize tail-risk objectives such as CVaR through sampling-based gradients.	Justifies explicit downside control as a design principle for RL rewards.
Hurst, Johnson, & Ooi (2010)	Explains why capital diversification and risk diversification can differ sharply in practice.	Motivates comparing RL with transparent risk-based benchmark portfolios.

Taken together, these papers support three core design choices in our project. First, they justify modeling allocation as a sequential decision problem rather than a one-shot optimization problem. Second, they motivate explicit risk control inside the learning objective instead of evaluating risk only after training. Third, they reinforce the importance

of comparing RL against transparent classical baselines and of treating validation design as part of the modeling contribution rather than as an afterthought.

Selected Reward Parameters

Table 5: Reward parameters selected on the validation window and used unchanged on the test set.

Parameter	Value
transaction_cost	0.0005
noise_sigma	0.1000
max_weight	0.7000
lambda	10.0000
phi	0.0200
eta	0.0005

Top Validation Configurations

Table 6: Top five reward-coefficient settings ranked by validation score.

lambda	phi	eta	Sharpe	Max DD	Validation Score
10	0.020	5e-04	0.51	-28.2%	0.373
10	0.005	1e-03	0.51	-33.6%	0.347
10	0.010	5e-04	0.49	-29.3%	0.340
10	0.010	1e-03	0.44	-30.4%	0.285
30	0.010	5e-04	0.43	-30.7%	0.276

COVID Stress-Test Summary

Table 7: Supplementary COVID-window stress test, February 19, 2020 to June 30, 2020.

Strategy	Total Return	Sharpe	Max DD
RL	1.5%	0.36	-33.3%
Buy-Hold EW	0.3%	0.19	-25.6%
Monthly EW	1.5%	0.29	-25.6%
Inverse Vol	-1.9%	-0.00	-25.6%
Min Variance	-3.2%	-0.12	-25.6%

Homework 2 Bibliography

Wang, X., & Liu, L. (2025). Risk-sensitive deep reinforcement learning for portfolio optimization. *Journal of Risk and Financial Management*, 18(7), 347.

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Tamar, A., Glassner, Y., & Mannor, S. (2014). Optimizing the CVaR via sampling. *arXiv:1404.3862*.

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